QUT Centre for Robotics



# ENN583 – Foundations of Robotic Vision

Week 11 – Sem 2 – 2026

Prof Niko Suenderhauf

# Welcome to Week 11!




CRICOS No.00213J

# Semantic Segmentation and Instance Segmentation

# Semantic Segmentation – Every Pixel gets a Class Label



- **Input:**
  - RGB Image
- **Output:**
  - One semantic class per pixel, such as floor, robot, box or person
- **Supervision:**
  - Pixel-level semantic mask
- **Loss (example):**
  - Pixel-wise cross-entropy
- **Useful Metric:**
  - Mean Intersection-over-Union (IoU)

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# Semantic Segmentation Output

- A vector of C logits per pixel
- What **activation** to use at the output?
  - Mutually exclusive classes → Softmax
  - Not exclusive classes → element-wise sigmoid

- Prediction: $\arg\max_c p(c \mid i, j)$



Predictions:
C x H x W

# Instance Segmentation – Separate Masks for Individual Object Instances



- **Input:**
  - RGB Image
- **Output:**
  - A separate pixel mask and class label for every object instance
- **Supervision:**
  - Individual mask and class label for each object
- **Loss (example):**
  - Classification, and mask losses
- **Useful Metric:**
  - Mask average precision

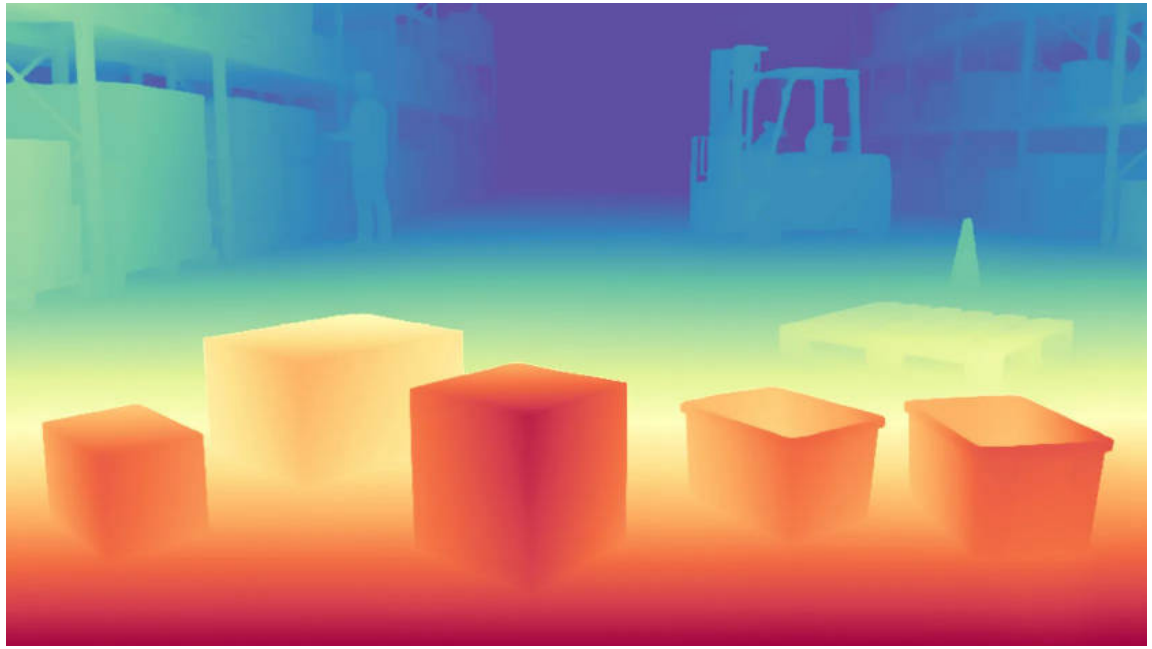
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# Semantic vs Instance Segmentation





# Architecturally Related: Depth Prediction and Other “Dense” Prediction Tasks



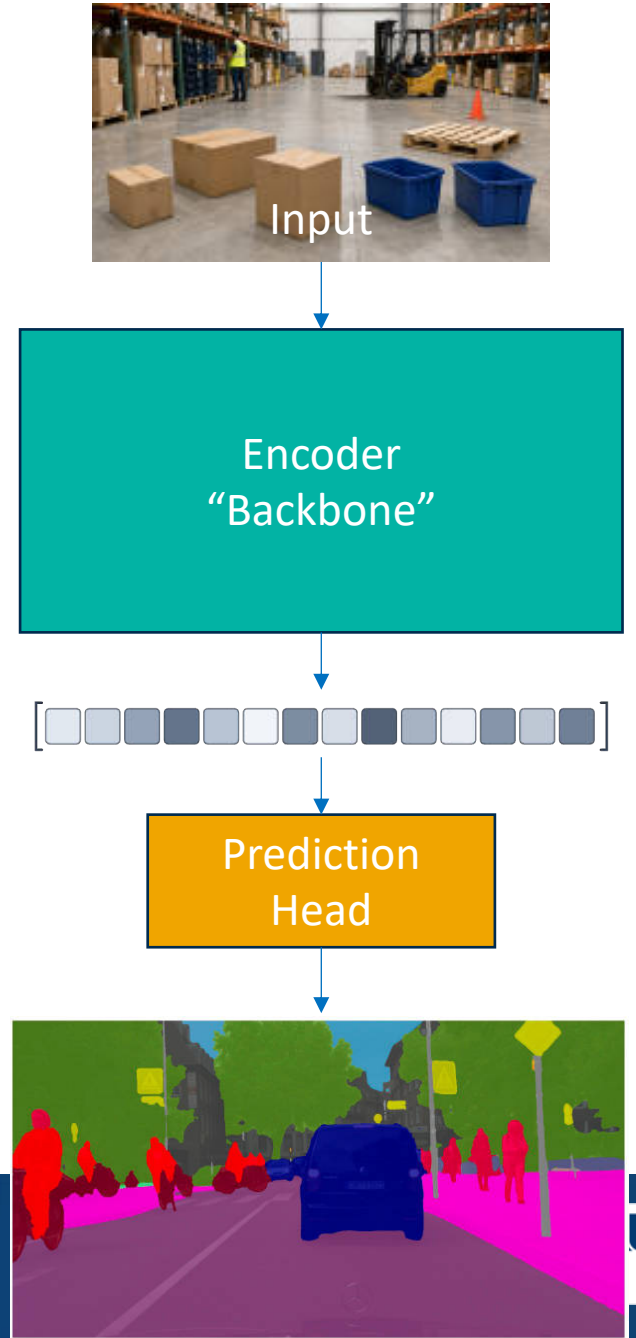
- **Input:**
  - RGB Image
- **Output:**
  - Estimated depth for every pixel
- **Supervision:**
  - Ground-truth depth map, e.g. from depth sensor or stereo system
- **Loss (example):**
  - L2 depth loss
- **Useful Metric:**
  - RMSE

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# How to Implement This?

**Problem:**

- output size = input size
- But the extracted feature after the encoder typically has smaller spatial dimensions



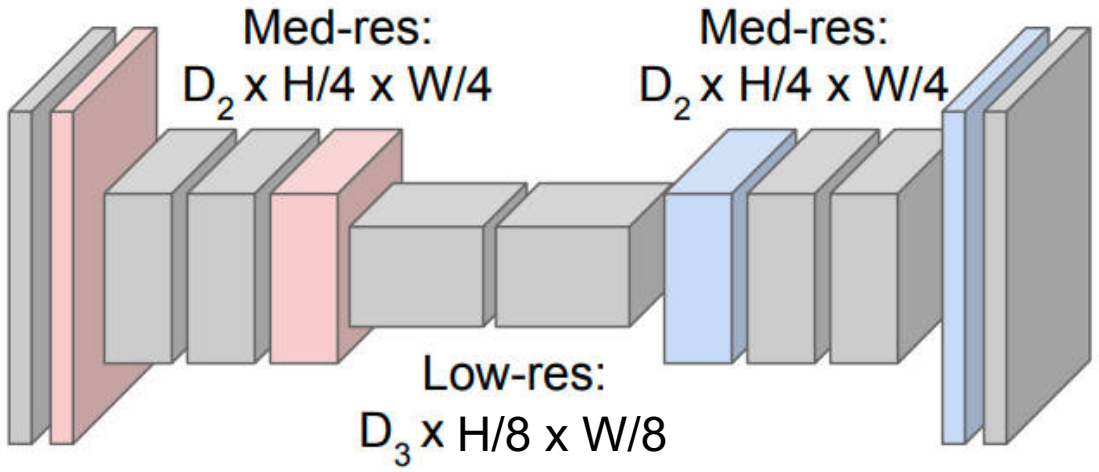
# Encoder – Decoder Architecture



Input:
3 x H x W

**Encoder**
downsampling

**Decoder**
upsampling




High-res:
$D_1$ x H/2 x W/2

High-res:
$D_1$ x H/2 x W/2



Predictions:
C x H x W



Slide courtesy Dr Dimity Miller, adapted from Fei-Fei Li, Justin Johnson and Serena Yeung

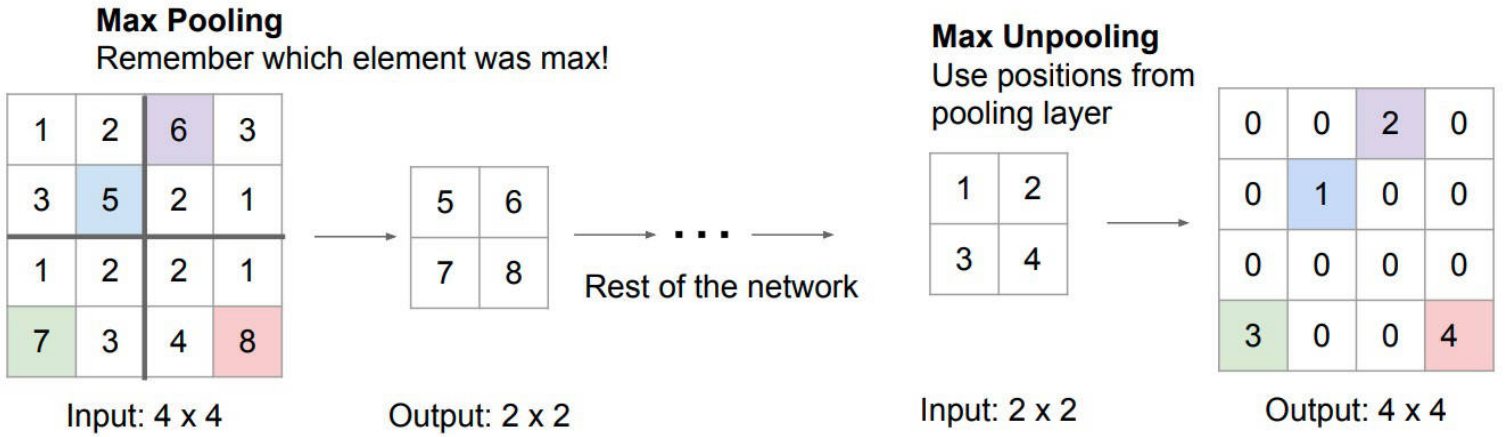
# Downsampling and Upsampling?

**Downsampling** (reduce spatial resolution)

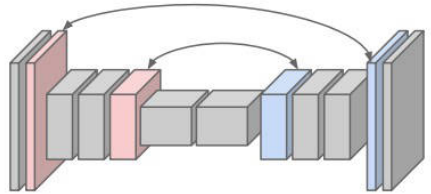
- Pooling Layers
- Strided Convolutions
- (see CAB420)

**Upsampling** (increase spatial resolution)

- Unpooling Layers



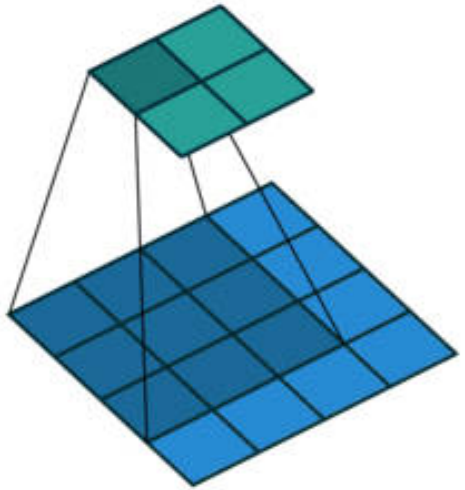

Corresponding pairs of downsampling and upsampling layers



# Downsampling and Upsampling?

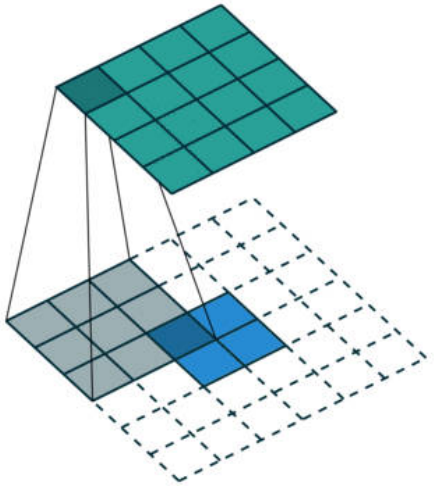
**Downsampling** (reduce spatial resolution)

- Pooling Layers
- Strided Convolutions
- (see CAB420)



**Upsampling** (increase spatial resolution)

- Unpooling Layers
- Transposed Convolutions



# Skip Connections and U-Net Architecture

- **Skip connections** between layers in the encoder and decoder
- Copy high-resolution (but semantically low-level) features from encoder to the decoder
- Helps preserve object boundaries and finer details



arXiv:1505.04597v1 [cs.CV] 18 May 2015

**U-Net: Convolutional Networks for Biomedical Image Segmentation**

Olaf Ronneberger, Philipp Fischer, and Thomas Brox

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ronneber@informatik.uni-freiburg.de,
WWW home page: http://lmb.informatik.uni-freiburg.de/

**Abstract.** There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding-window convolutional network) on the ISBI challenge for segmentation of neuronal structures in electron microscopic stacks. Using the same network trained on transmitted light microscopy images (phase contrast and DIC) we won the ISBI cell tracking challenge 2015 in these categories by a large margin. Moreover, the network is fast. Segmentation of a 512x512 image takes less than a second on a recent GPU. The full implementation (based on Caffe) and the trained networks are available at http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net.

**1 Introduction**

In the last two years, deep convolutional networks have outperformed the state of the art in many visual recognition tasks, e.g. [7,3]. While convolutional networks have already existed for a long time [8], their success was limited due to the size of the available training sets and the size of the considered networks. The breakthrough by Krizhevsky et al. [7] was due to supervised training of a large network with 8 layers and millions of parameters on the ImageNet dataset with 1 million training images. Since then, even larger and deeper networks have been trained [12].

The typical use of convolutional networks is on classification tasks, where the output to an image is a single class label. However, in many visual tasks, especially in biomedical image processing, the desired output should include localization, i.e., a class label is supposed to be assigned to each pixel. Moreover, thousands of training images are usually beyond reach in biomedical tasks. Hence, Ciresan et al. [1] trained a network in a sliding-window setup to predict the class label of each pixel by providing a local region (patch) around that pixel

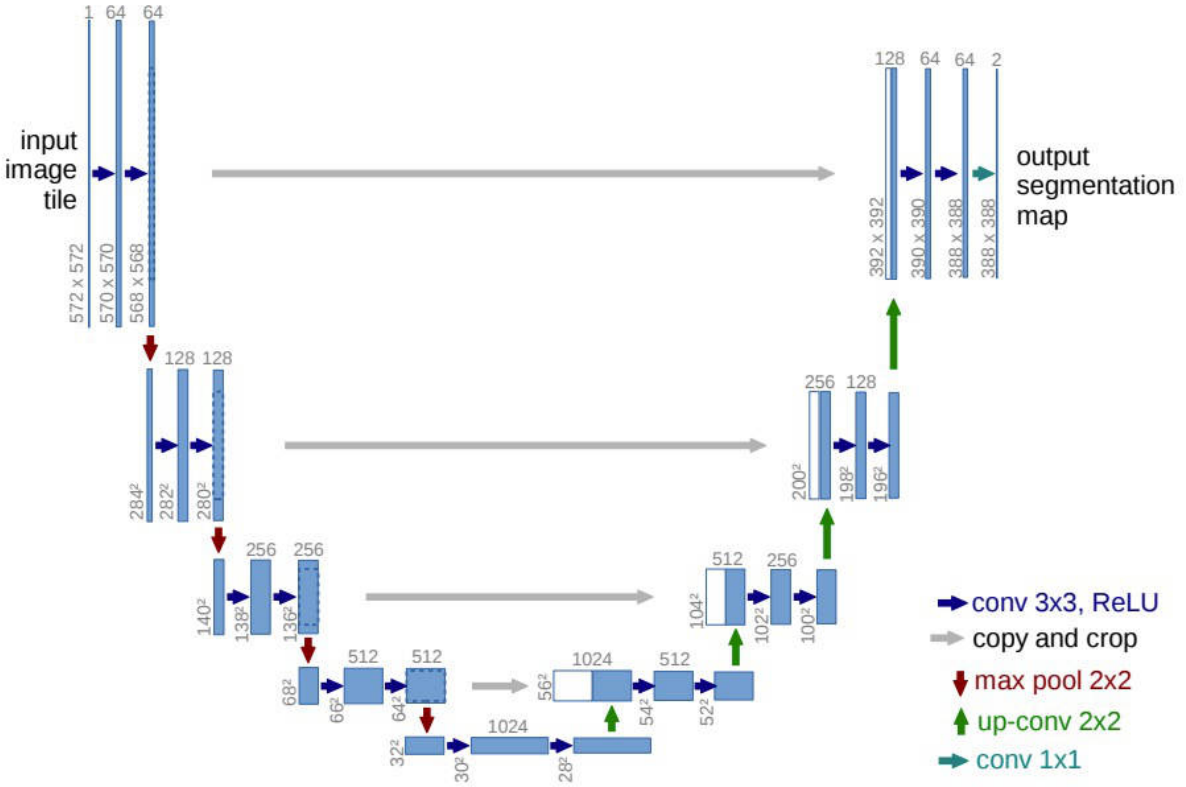



**Fig. 1.** U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

## Encoder (contracting path)

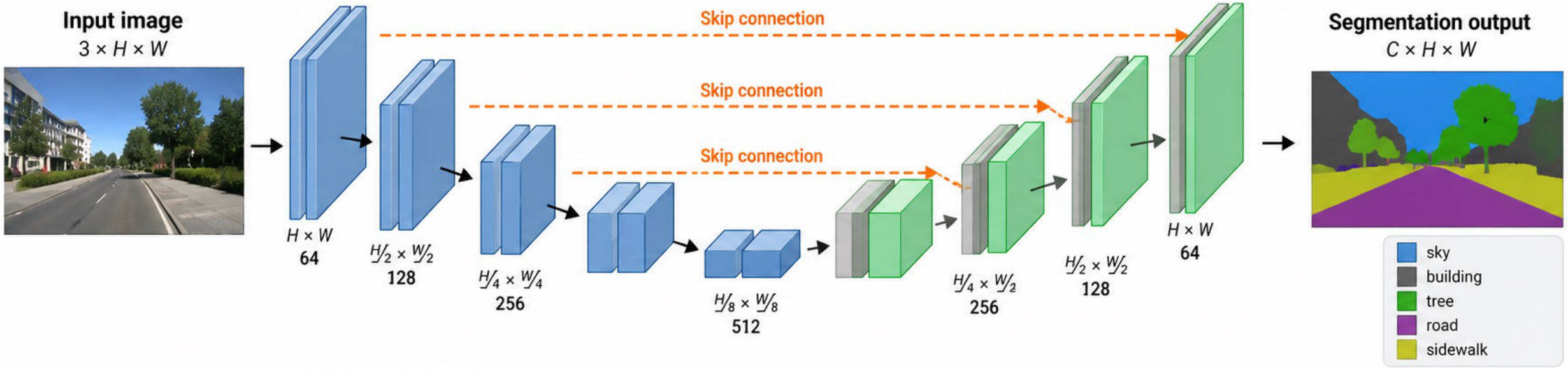
- Repeated convolutions and downsampling
- Increases the receptive field (more context)
- Lower spatial resolution, higher-level features

## Skip connections

- Copy high-resolution features from encoder to decoder
- Help recover fine spatial detail and preserve object boundaries

## Decoder (expanding path)

- Repeated **upsampling** and convolutions
- Combines high-level context (from encoder) with high-resolution details (via skip connections)
- Produces dense per-pixel class predictions




**Feature map examples** (same image location)



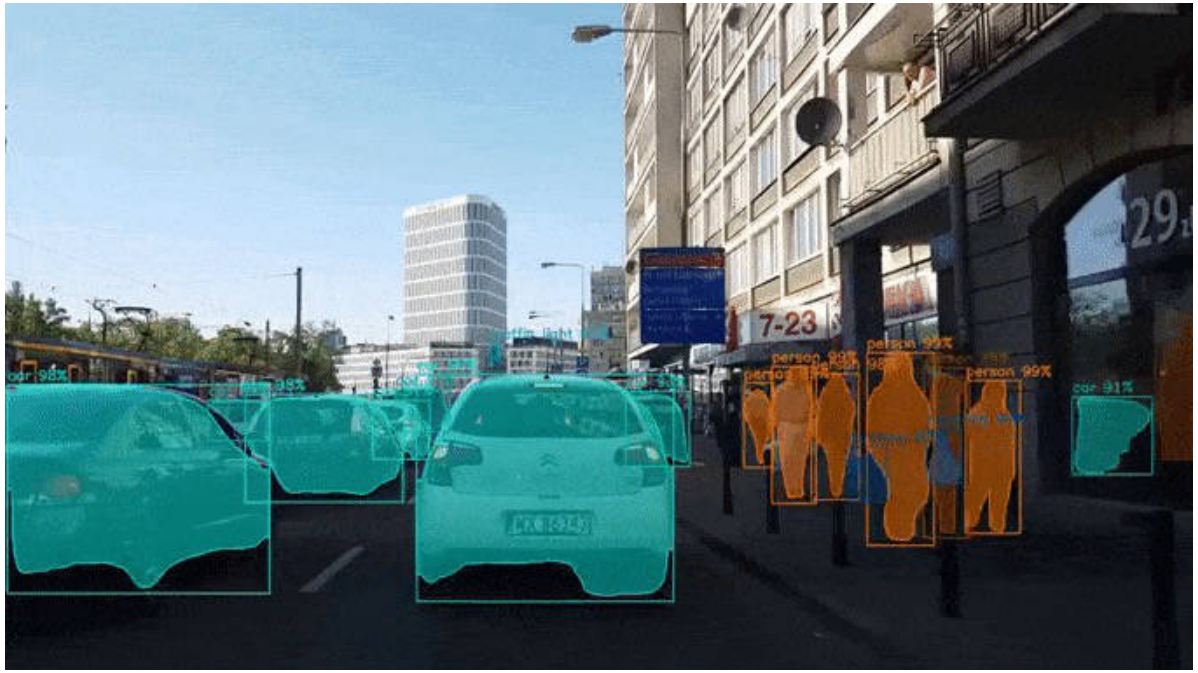

Early encoder feature
(high resolution)
Edges and texture

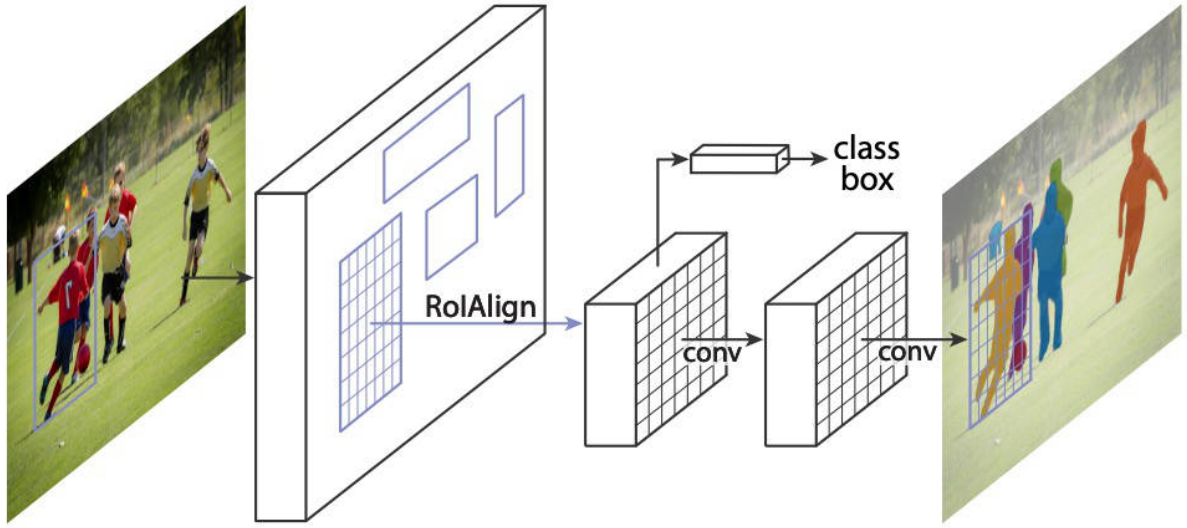
Deeper encoder feature
(low resolution)
High-level semantic context

Decoder feature
(upsampled and combined)
Recovers object boundaries

Final prediction
Per-pixel class probabilities
(softmax over C classes)

# Instance Segmentation: Mask R-CNN






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# Promptable Segmentation: Segment Anything

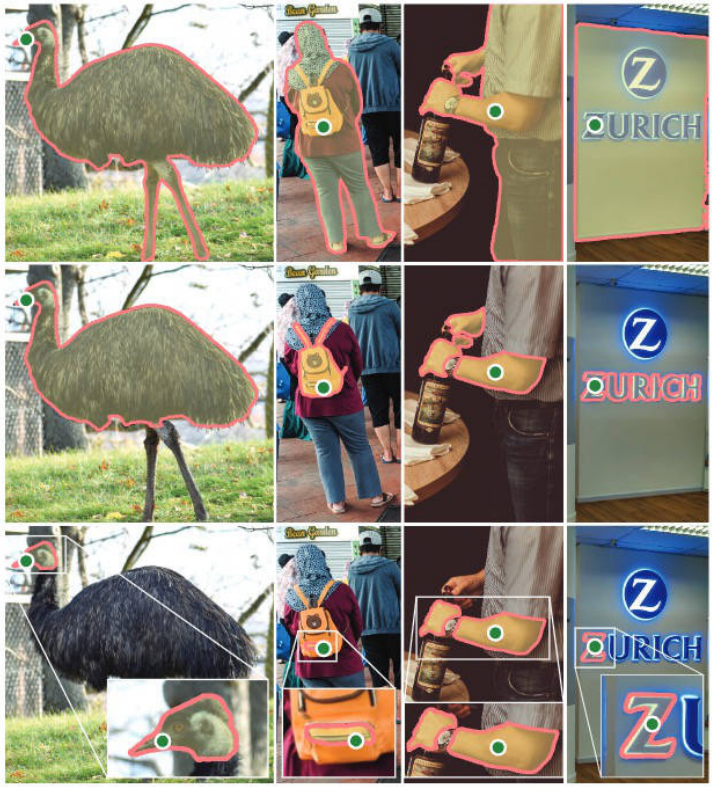


Figure 3: Each column shows 3 valid masks generated by SAM from a single ambiguous point prompt (green circle).

arXiv:2408.00714v2 [cs.CV] 28 Oct 2024

## SAM 2: Segment Anything in Images and Videos

Nikhila Ravi*,†, Valentin Gabeur*, Yuan-Ting Hu*, Ronghang Hu*, Chaitanya Ryali*, Tengyu Ma*, Haitham Khedr*, Roman Rädle*, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár†, Christoph Feichtenhofer*,†

Meta FAIR

*core contributor, †project lead

We present Segment Anything Model 2 (SAM 2), a foundation model towards solving promptable visual segmentation in images and *videos*. We build a data engine, which improves model and data via user interaction, to collect the largest video segmentation dataset to date. Our model is a simple transformer architecture with streaming memory for real-time video processing. SAM 2 trained on our data provides strong performance across a wide range of tasks. In video segmentation, we observe better accuracy, using 3× fewer interactions than prior approaches. In image segmentation, our model is more accurate and 6× faster than the Segment Anything Model (SAM). We believe that our data, model, and insights will serve as a significant milestone for video segmentation and related perception tasks. We are releasing our main model, dataset, as well as code for model training and our demo.

**Demo:** https://sam2.metademolab.com
**Code:** https://github.com/facebookresearch/sam2
**Website:** https://ai.meta.com/sam2

Meta

### 1 Introduction

Segment Anything (SA) introduced a foundation model for promptable *segmentation* in *images* (Kirillov et al., 2023). However an image is only a static snapshot of the real world in which visual segments can exhibit complex motion, and with the rapid growth of multimedia content, a significant portion is now recorded with a temporal dimension, particularly in *video* data. Many important applications in AR/VR, robotics, autonomous vehicles, and video editing require temporal localization beyond image-level segmentation. We believe a universal visual segmentation system should be applicable to both images *and* videos.

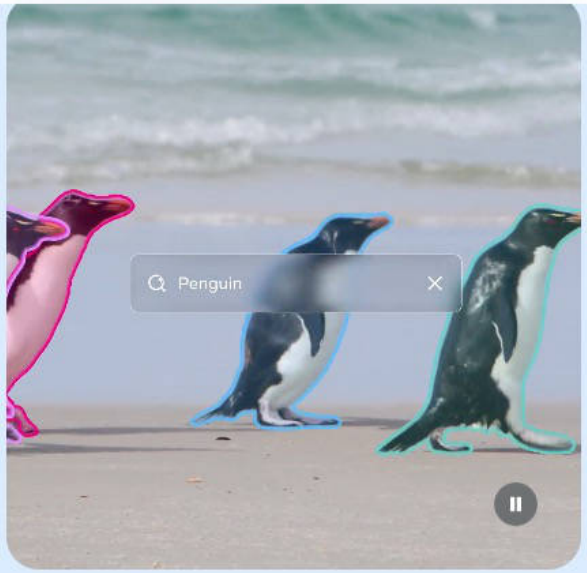
Segmentation in video aims to determine the spatio-temporal extent of entities, which presents unique challenges beyond those in images. Entities can undergo significant changes in appearance due to motion, deformation, occlusion, lighting changes, and other factors. Videos often have lower quality than images due to camera motion, blur, and lower resolution. Further, efficient processing of a large number of frames is a key challenge. While SA successfully addresses segmentation in images, existing video segmentation models and datasets fall short in providing a comparable capability to "segment *anything* in videos".

We introduce the Segment Anything Model 2 (SAM 2), a *unified* model for video and image segmentation (we consider an image as a single-frame video). Our work includes a task, model, and dataset (see Fig. 1).

We focus on the Promptable Visual Segmentation (PVS) *task* that generalizes image segmentation to the video domain. The task takes as input points, boxes, or masks on any frame of the video to define a segment of interest for which the spatio-temporal mask (i.e., a '*masklet*') is to be predicted. Once a masklet is predicted, it can be iteratively refined by providing prompts in additional frames.

Our *model* (§4) produces segmentation masks of the object of interest, in single images *and* across video frames. SAM 2 is equipped with a memory that stores information about the object and previous interactions, which allows it to generate masklet predictions throughout the video, and also effectively correct these based on the stored memory context of the object from previously observed frames. Our streaming architecture is a natural generalization of SAM to the video domain, processing video frames one at a time, equipped with a memory attention module to attend to the previous memories of the target object. When applied to images, the memory is empty and the model behaves like SAM.

# Open Vocabulary Segmentation – SAM3




## SAM 3

**Detect, segment and track every example of any object category in an image or video, using text or examples**

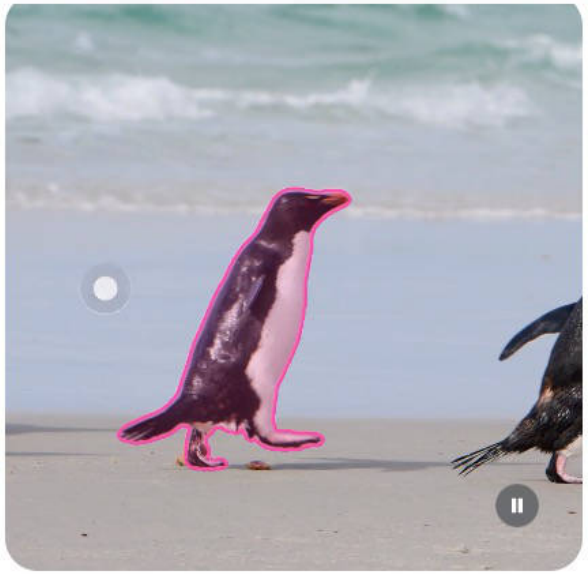
- Segment an object from a click
- Track segmented objects in videos
- Refine prediction with follow up clicks
- **Detect and segment matching instances from text**
- **Refine detection with visual examples**



## SAM 2

**Segment and track any object in any image or video using click, box or mask prompts**

- Segment an object from a click
- Track segmented objects in videos
- Refine prediction with follow up clicks



## SAM 1

**Segment any object in any image with as little as a single click**

- Segment an object from a click
- Refine prediction with follow up clicks

# Modern Segmentation Models

**Modern CNN-based:**

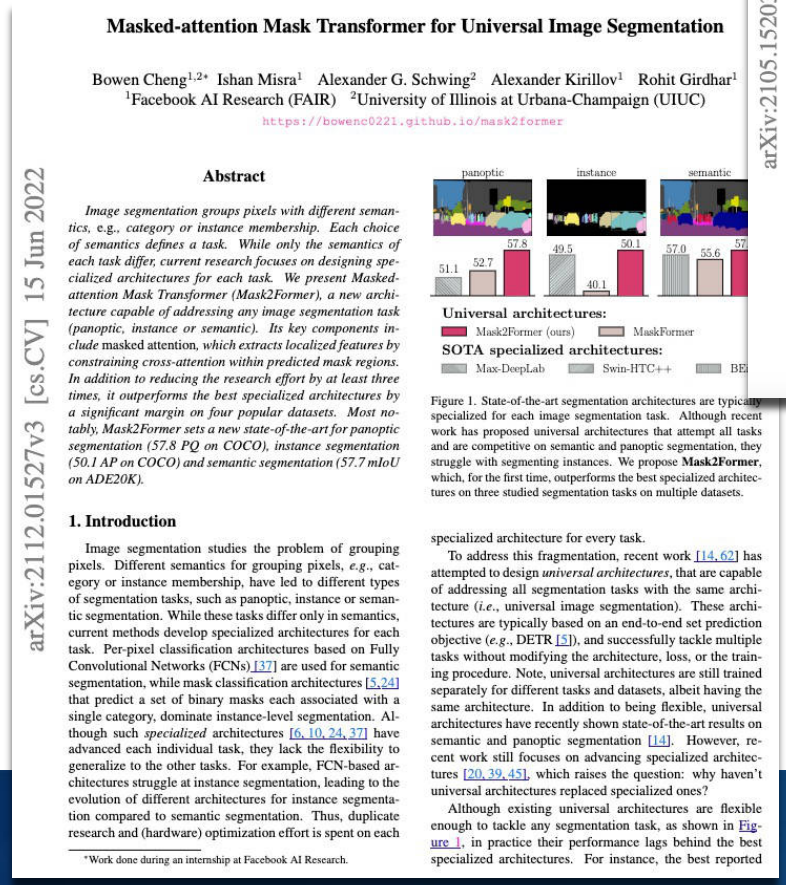
- DeepLabv3, DeeLabv3+

**Transformer-based:**

- Mask2Former
- SegFormer

**Promptable / Foundation Models**

- SAM2, SAM3



arXiv:2112.01527v3 [cs.CV] 15 Jun 2022

**Masked-attention Mask Transformer for Universal Image Segmentation**

Bowen Cheng[1,2*] Ishan Misra[1] Alexander G. Schwing[2] Alexander Kirillov[1] Rohit Girdhar[1]
[1]Facebook AI Research (FAIR) [2]University of Illinois at Urbana-Champaign (UIUC)

https://bowenc0221.github.io/mask2former

**Abstract**

*Image segmentation groups pixels with different semantics, e.g., category or instance membership. Each choice of semantics defines a task. While only the semantics of each task differ, current research focuses on designing specialized architectures for each task. We present Masked-attention Mask Transformer (Mask2Former), a new architecture capable of addressing any image segmentation task (panoptic, instance or semantic). Its key components include* masked attention, *which extracts localized features by constraining cross-attention within predicted mask regions. In addition to reducing the research effort by at least three times, it outperforms the best specialized architectures by a significant margin on four popular datasets. Most notably, Mask2Former sets a new state-of-the-art for panoptic segmentation (57.8 PQ on COCO), instance segmentation (50.1 AP on COCO) and semantic segmentation (57.7 mIoU on ADE20K).*

**1. Introduction**

Image segmentation studies the problem of grouping pixels. Different semantics for grouping pixels, *e.g.*, category or instance membership, have led to different types of segmentation tasks, such as panoptic, instance or semantic segmentation. While these tasks differ only in semantics, current methods develop specialized architectures for each task. Per-pixel classification architectures based on Fully Convolutional Networks (FCNs) [37] are used for semantic segmentation, while mask classification architectures [5, 24] that predict a set of binary masks each associated with a single category, dominate instance-level segmentation. Although such *specialized* architectures [6, 10, 24, 37] have advanced each individual task, they lack the flexibility to generalize to the other tasks. For example, FCN-based architectures struggle at instance segmentation, leading to the evolution of different architectures for instance segmentation compared to semantic segmentation. Thus, duplicate research and (hardware) optimization effort is spent on each specialized architecture for every task.

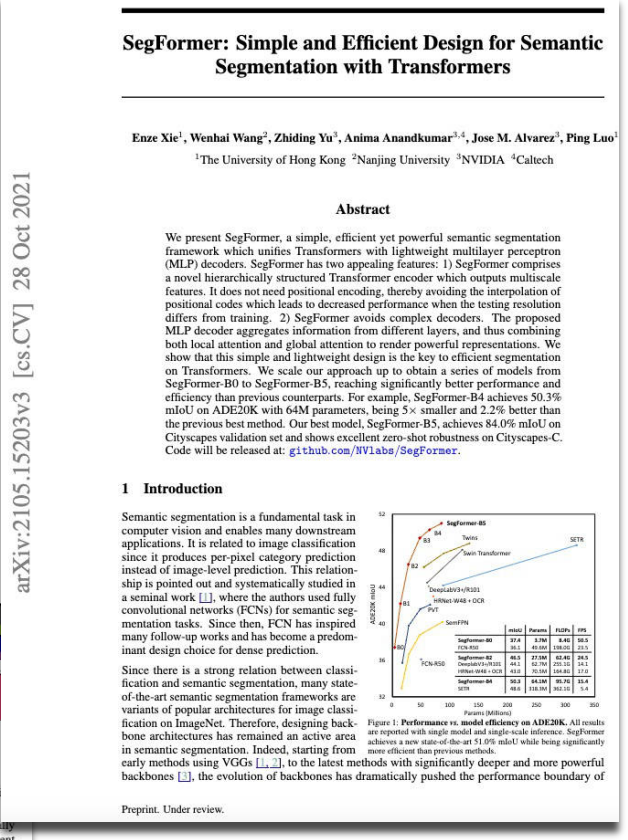
To address this fragmentation, recent work [14, 62] has attempted to design *universal architectures*, that are capable of addressing all segmentation tasks with the same architecture (*i.e.*, universal image segmentation). These architectures are typically based on an end-to-end set prediction objective (*e.g.*, DETR [5]), and successfully tackle multiple tasks without modifying the architecture, loss, or the training procedure. Note, universal architectures are still trained separately for different tasks and datasets, albeit having the same architecture. In addition to being flexible, universal architectures have recently shown state-of-the-art results on semantic and panoptic segmentation [14]. However, recent work still focuses on advancing specialized architectures [20, 39, 45], which raises the question: why haven't universal architectures replaced specialized ones?

Although existing universal architectures are flexible enough to tackle any segmentation task, as shown in Figure 1, in practice their performance lags behind the best specialized architectures. For instance, the best reported

*Work done during an internship at Facebook AI Research.




Figure 1. State-of-the-art segmentation architectures are typically specialized for each image segmentation task. Although recent work has proposed universal architectures that attempt all tasks and are competitive on semantic and panoptic segmentation, they struggle with segmenting instances. We propose **Mask2Former**, which, for the first time, outperforms the best specialized architectures on three studied segmentation tasks on multiple datasets.



arXiv:2105.15203v3 [cs.CV] 28 Oct 2021

**SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers**

Enze Xie[1], Wenhai Wang[2], Zhiding Yu[3], Anima Anandkumar[3,4], Jose M. Alvarez[3], Ping Luo[1]
[1]The University of Hong Kong [2]Nanjing University [3]NVIDIA [4]Caltech

**Abstract**

We present SegFormer, a simple, efficient yet powerful semantic segmentation framework which unifies Transformers with lightweight multilayer perceptron (MLP) decoders. SegFormer has two appealing features: 1) SegFormer comprises a novel hierarchically structured Transformer encoder which outputs multiscale features. It does not need positional encoding, thereby avoiding the interpolation of positional codes which leads to decreased performance when the testing resolution differs from training. 2) SegFormer avoids complex decoders. The proposed MLP decoder aggregates information from different layers, and thus combining both local attention and global attention to render powerful representations. We show that this simple and lightweight design is the key to efficient segmentation on Transformers. We scale our approach up to obtain a series of models from SegFormer-B0 to SegFormer-B5, reaching significantly better performance and efficiency than previous counterparts. For example, SegFormer-B4 achieves 50.3% mIoU on ADE20K with 64M parameters, being 5× smaller and 2.2% better than the previous best method. Our best model, SegFormer-B5, achieves 84.0% mIoU on Cityscapes validation set and shows excellent zero-shot robustness on Cityscapes-C. Code will be released at: github.com/NVlabs/SegFormer.

**1 Introduction**

Semantic segmentation is a fundamental task in computer vision and enables many downstream applications. It is related to image classification since it produces per-pixel category prediction instead of image-level prediction. This relationship is pointed out and systematically studied in a seminal work [1], where the authors used fully convolutional networks (FCNs) for semantic segmentation tasks. Since then, FCN has inspired many follow-up works and has become a predominant design choice for dense prediction.

Since there is a strong relation between classification and semantic segmentation, many state-of-the-art semantic segmentation frameworks are variants of popular architectures for image classification on ImageNet. Therefore, designing backbone architectures has remained an active area in semantic segmentation. Indeed, starting from early methods using VGGs [1, 2], to the latest methods with significantly deeper and more powerful backbones [3], the evolution of backbones has dramatically pushed the performance boundary of

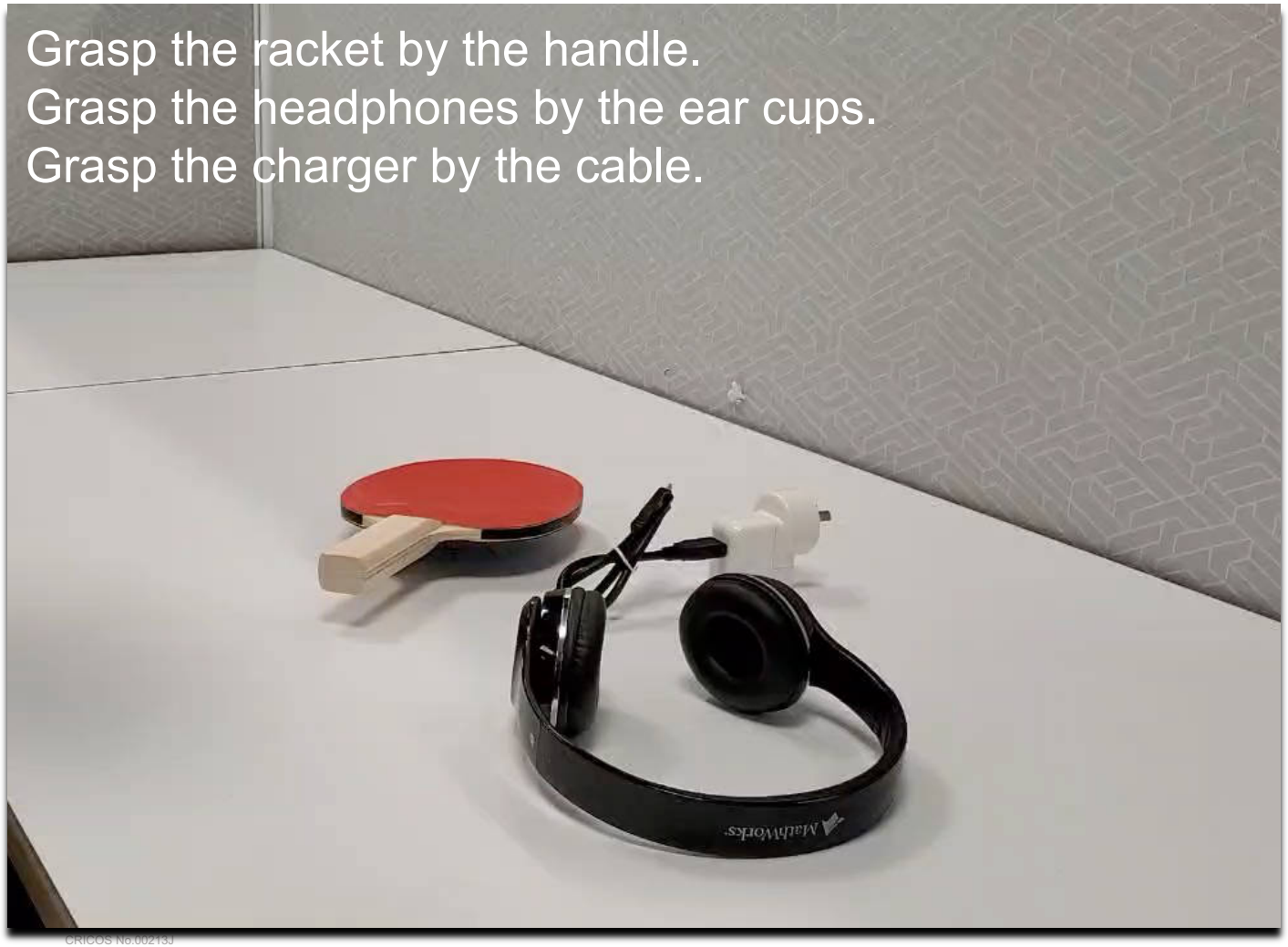


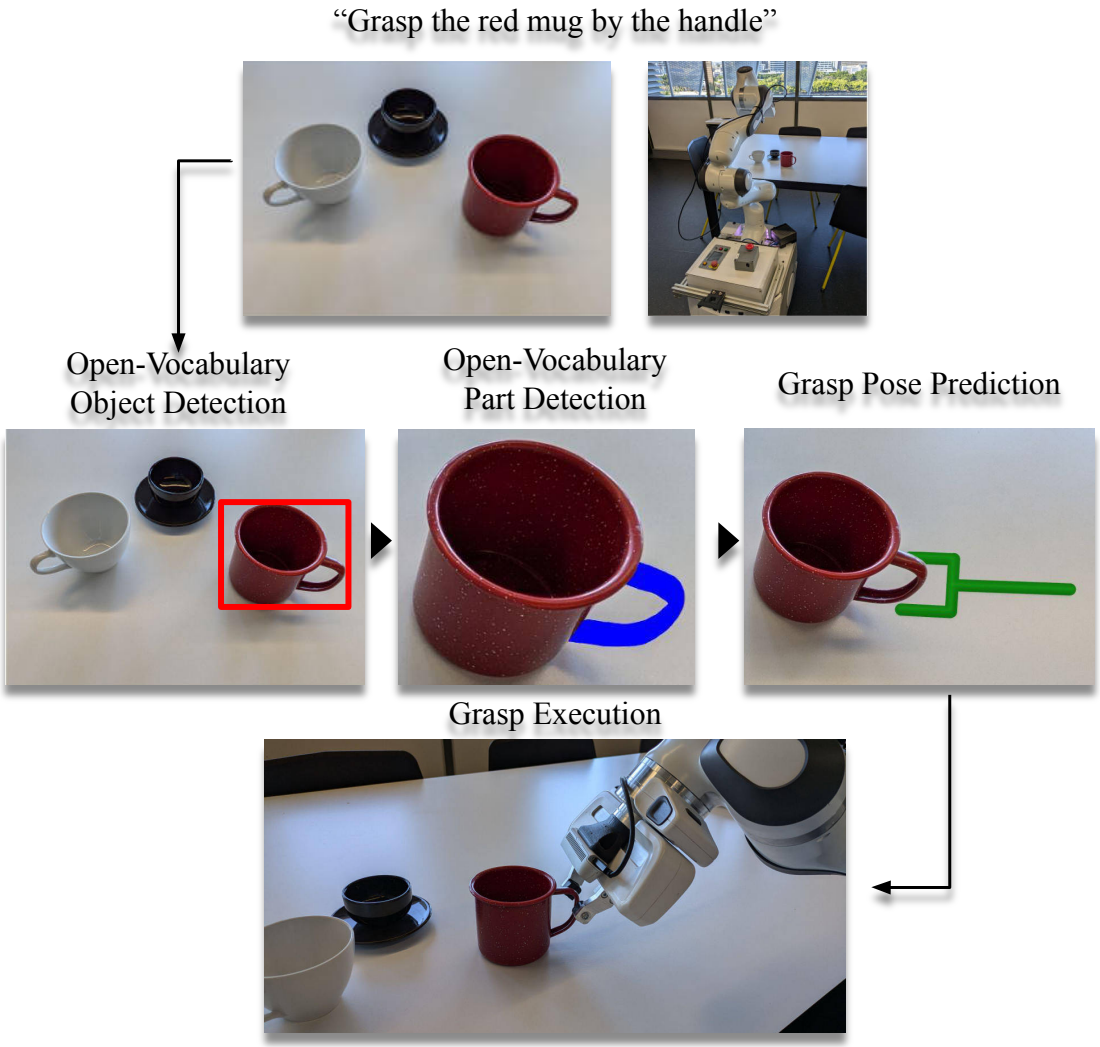

Figure 1: **Performance *vs.* model efficiency on ADE20K.** All results are reported with single model and single-scale inference. SegFormer achieves a new state-of-the-art 51.0% mIoU while being significantly more efficient than previous methods.

Preprint. Under review.

# Grasp Prediction / Synthesis

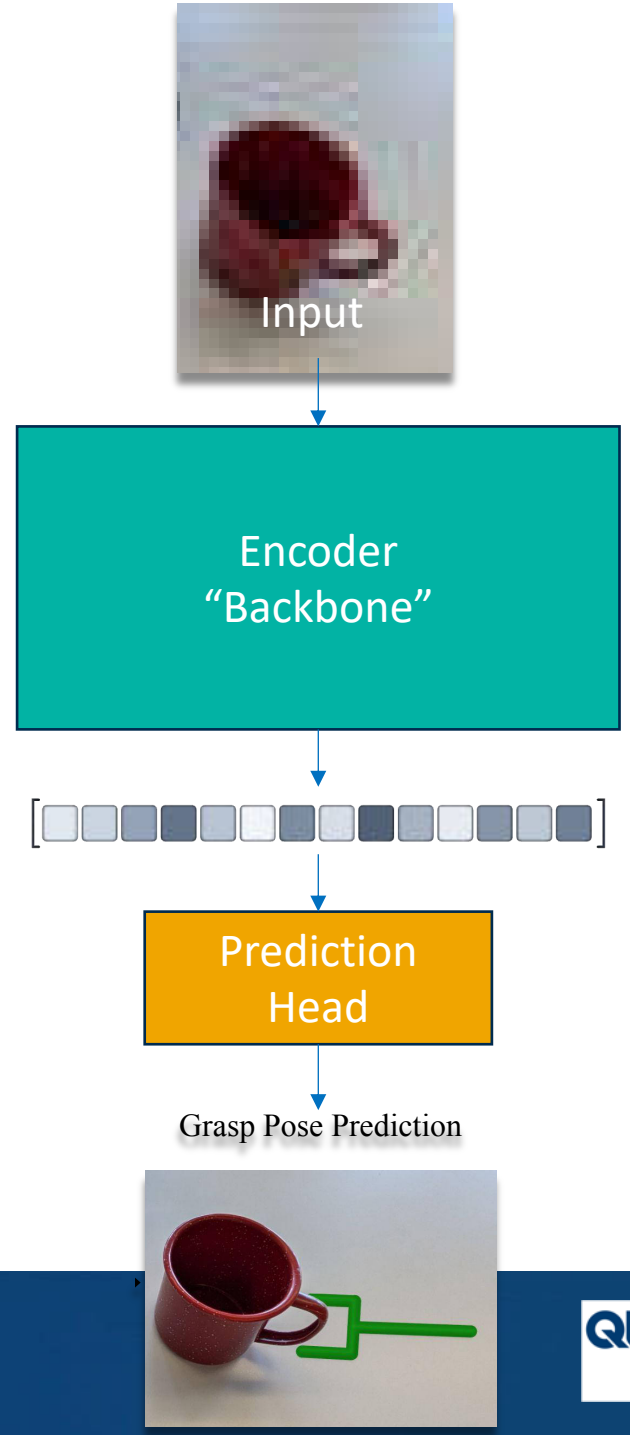
# Open-Vocabulary Part-Based Grasping



# What is Grasp Prediction?

Given an image of an object,
how should a robot pick-up this object?

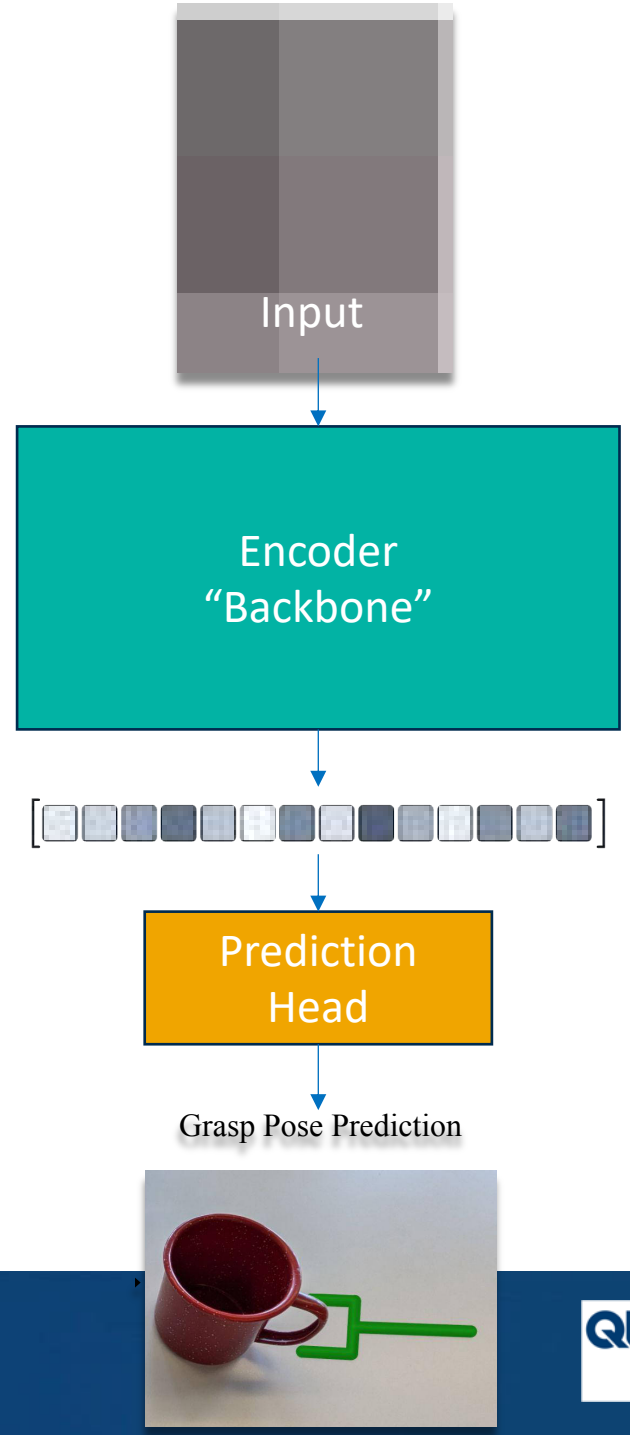


# What is Grasp Prediction?

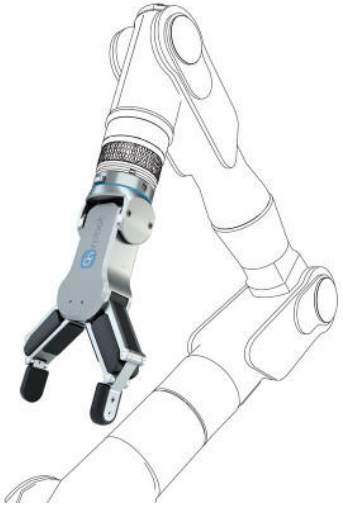
Given an image of an object,
how should a robot pick-up this object?

**Two categories:**

- Top-down grasping (simplified 2D grasps)
- 6-DoF Grasping (full 3D grasps)

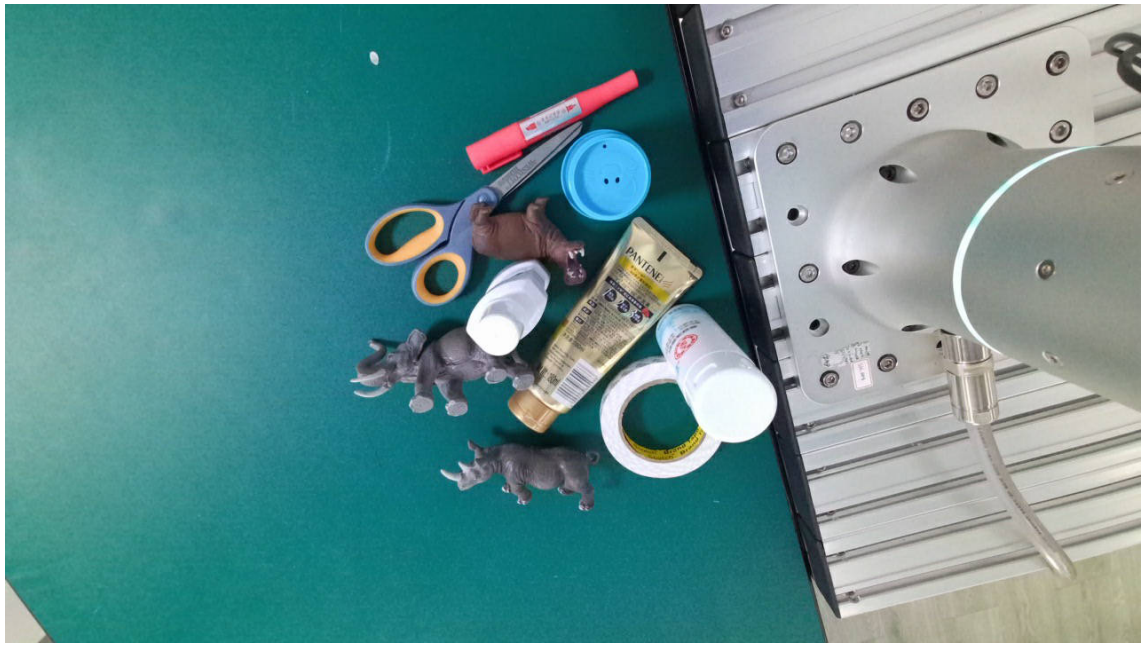


# What is Grasp Prediction?



**How to parametrise a grasp?**

- **Top-down grasps:** $(x, y, \theta, w, q)$
  - $x, y$ – grasp centre in the image
  - $\theta$ – gripper orientation in the image plane
  - $w$ – gripper opening needed for that grasp
  - $q$ – grasp quality / confidence score



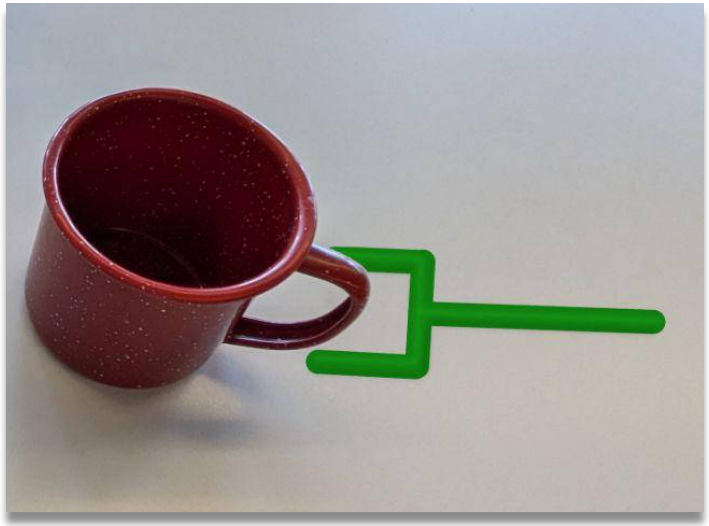
# What is Grasp Prediction?



**How to parametrise a grasp?**

- **6DOF Grasps:** $(x, y, z, \phi, \theta, \psi, w, q)$
  - $x, y, \mathrm{z}$ – grasp centre as 3D position
  - $\phi, \theta, \psi$ – 3D gripper orientation
  - $w$ – gripper opening needed for that grasp
  - $q$ – grasp quality / confidence score

  - 6 DoF = 3D position + 3D orientation

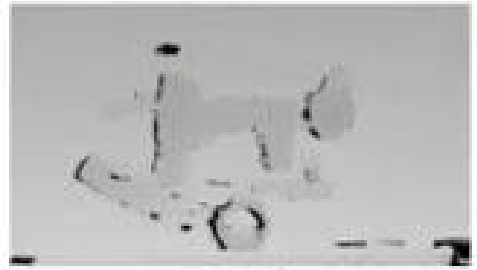


# Input Modalities

Depth image or point cloud in addition to RGB image are often used to provide important geometric information.



RGB



Depth

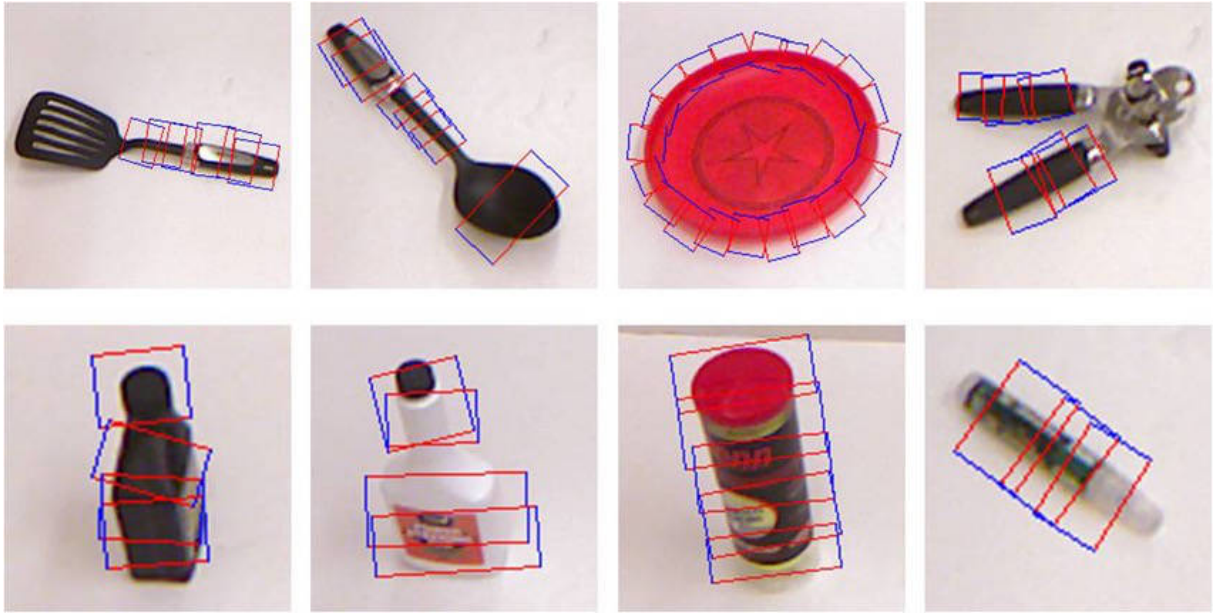


Point Clouds

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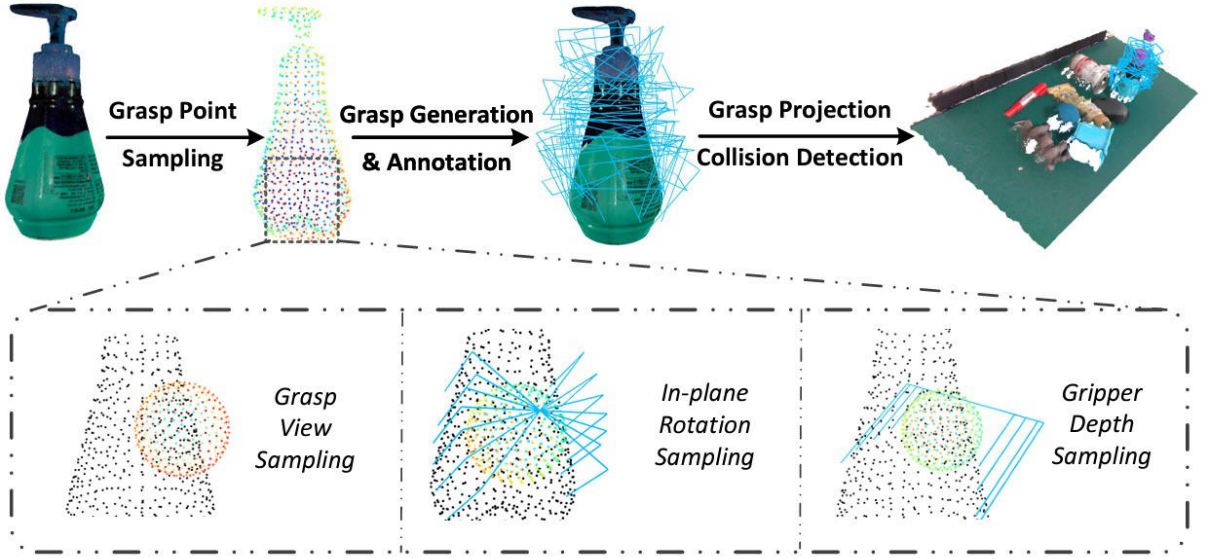
# Training Data

- Human labels (e.g. Cornell Grasp Dataset)



- Simulation (e.g. or Jacquard Dataset)

- Real RGB-D + analytically generated grasps (e.g. GraspNet-1 Billion)



# Two Approaches to Grasp Prediction

**Generative –** Given the visual data, generate a high-quality grasp



Model

$$(x, y, \theta, w, q)$$

Here “generative” means directly generating grasp candidates;
not necessarily a modern probabilistic generative model such as a diffusion model.

# Two Approaches to Grasp Prediction

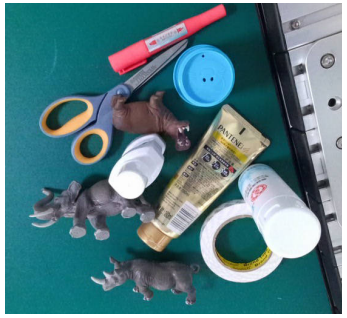
**Generative –** Given the visual data, generate a high-quality grasp



Model

$(x, y, \theta, w, q)$

**Discriminative –** Given the visual data and a proposed grasp, rate the quality of the grasp

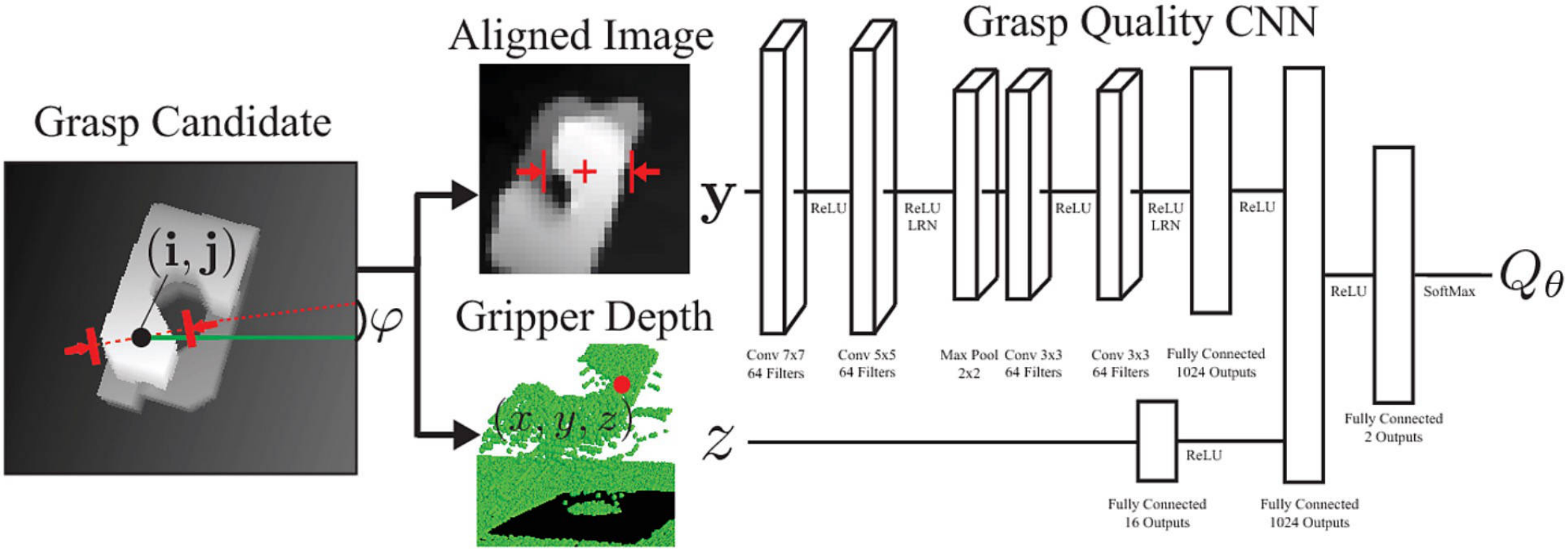


$(x, y, \theta, w)$

Model

90% chance of success

# Discriminative Grasp Prediction

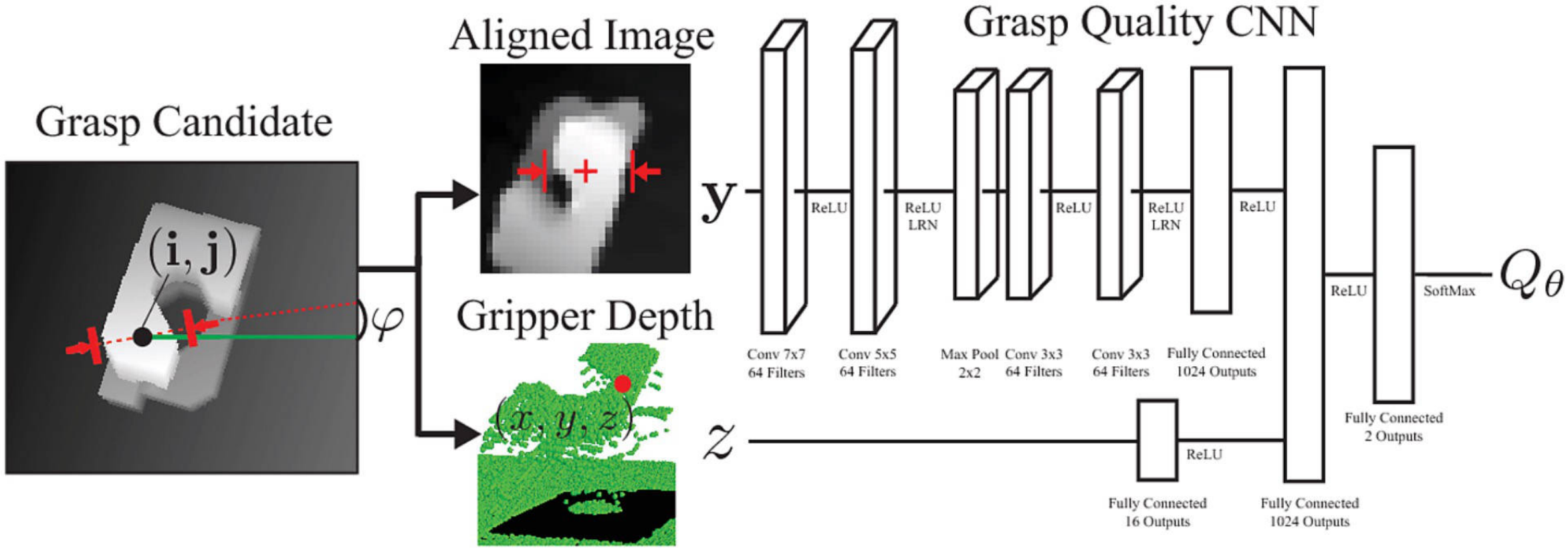





Mahler, Jeffrey, et al. "Dex-net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics." *arXiv preprint arXiv:1703.09312* (2017).

# Discriminative Grasp Prediction

Slow! Requires explicitly evaluating candidate grasps; computational cost grows with the number of candidates.



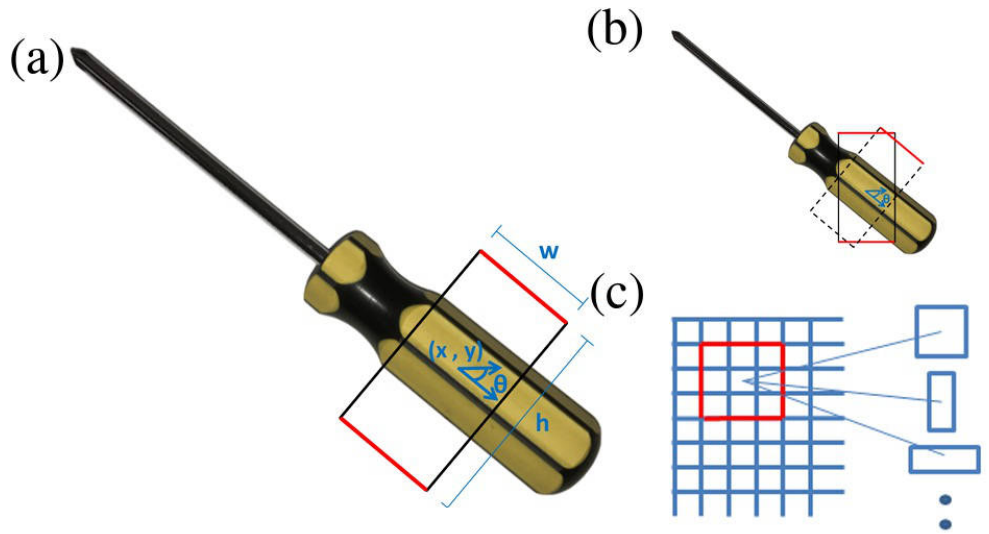


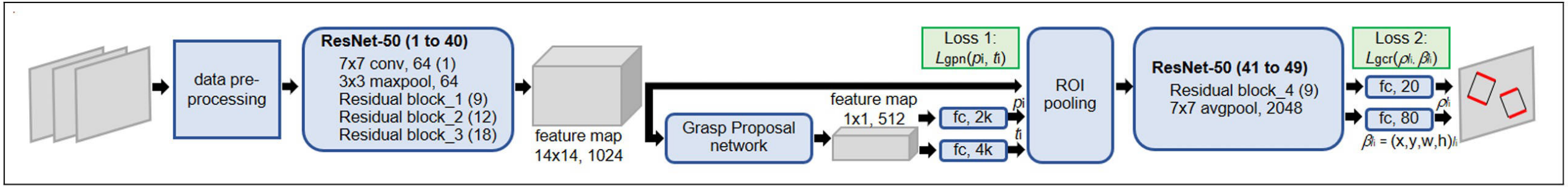

Mahler, Jeffrey, et al. "Dex-net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics." *arXiv preprint arXiv:1703.09312* (2017).

# Generative Grasp Prediction

**Multi-grasp:** Formulate as an object detection task!
Uses essentially a Faster R-CNN architecture.

Can generate multiple grasps per image in one go.



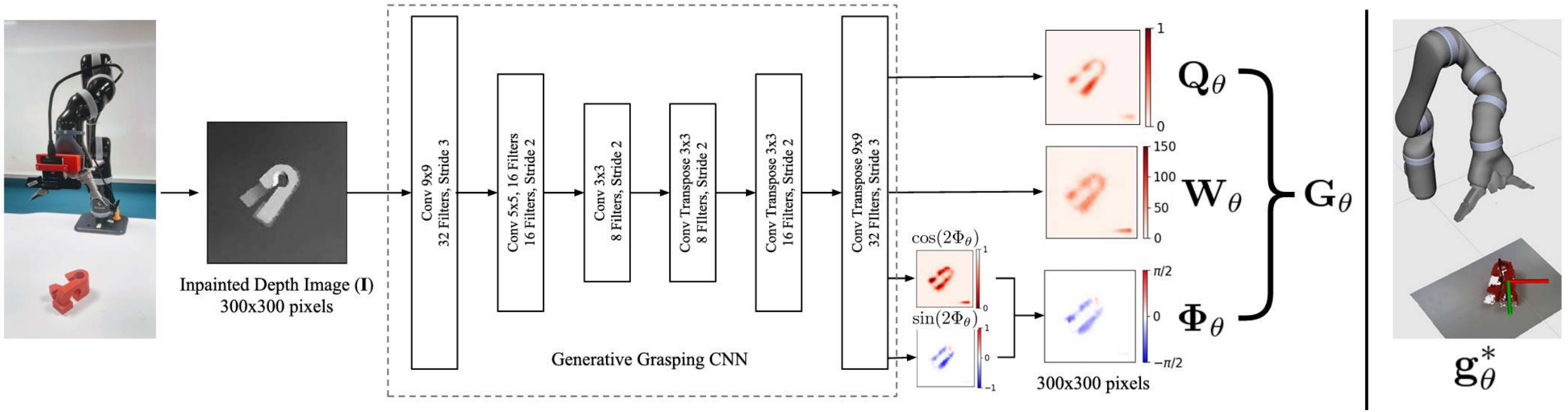



Chu, Fu-Jen, Ruinian Xu, and Patricio A. Vela. "Real-world multiobject, multigrasp detection." *IEEE Robotics and Automation Letters* 3.4 (2018): 3355-3362.

# Generative Grasp Prediction

**Generative Grasp CNN (GGCNN):** dense prediction, analogous to segmentation!

Predicts grasp angle, gripper width and quality for every pixel.

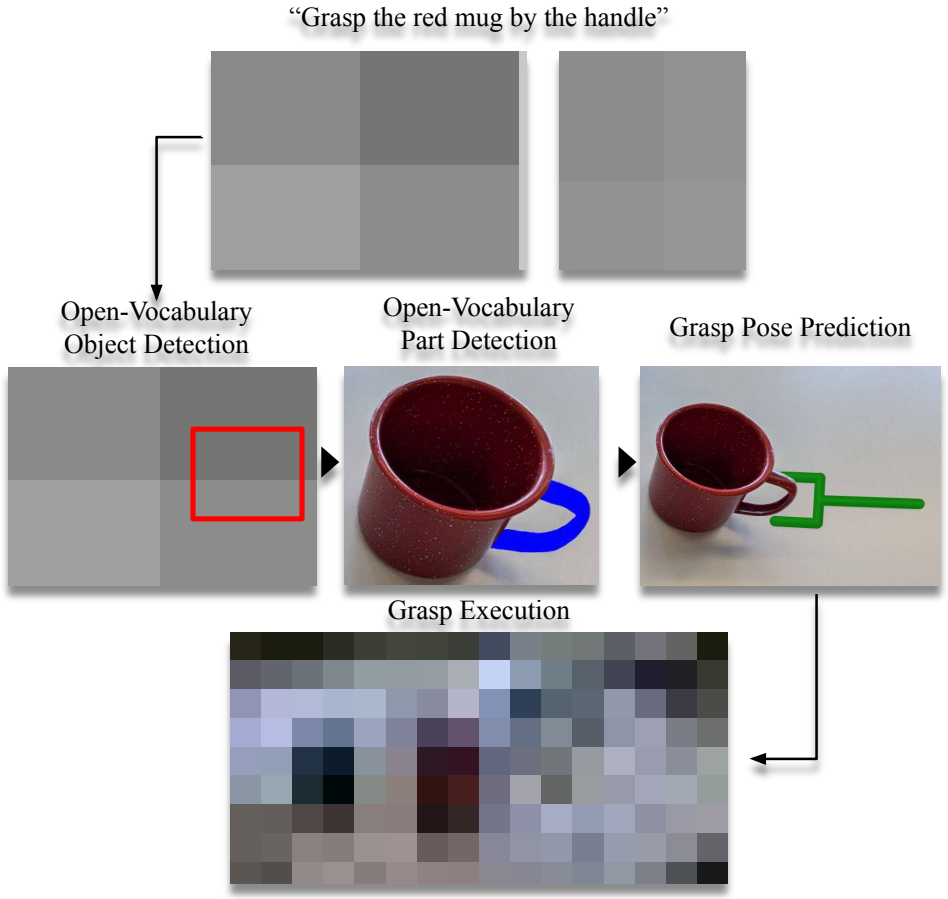





Morrison, Douglas, Peter Corke, and Jürgen Leitner. "Learning robust, real-time, reactive robotic grasping." *The International journal of robotics research* 39.2-3 (2020): 183-201.

# Top-Down Grasping is Limited!



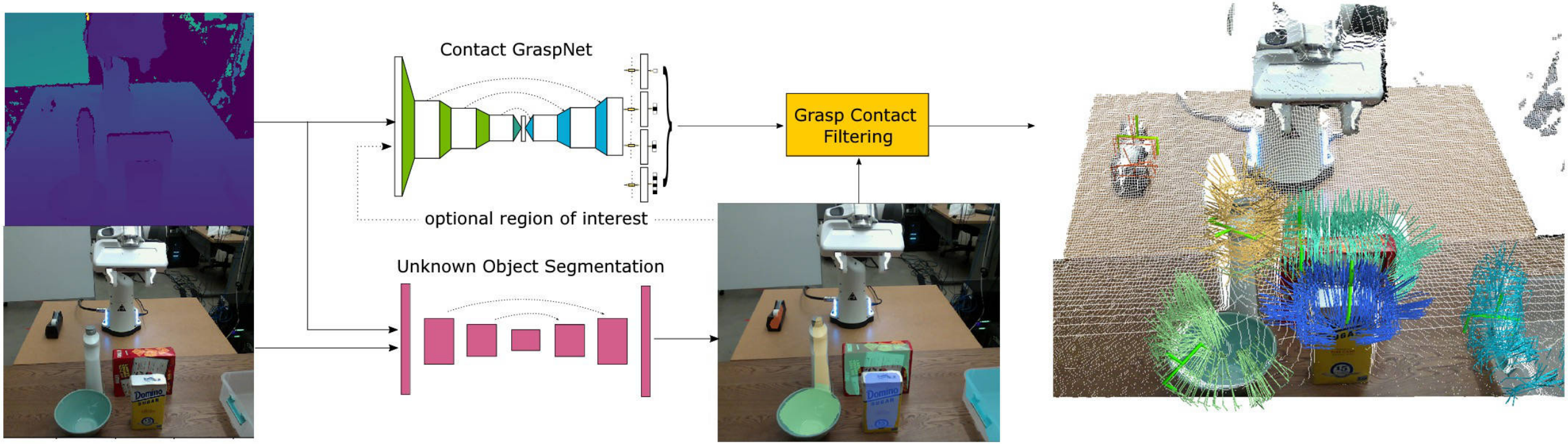



Top-down grasps assume a fixed approach direction and cannot exploit arbitrary 3D object surfaces.

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# 6-DoF Grasping: Contact-GraspNet

Idea: For each point in a 3D point cloud, predict whether it is a good grasp contact.
If yes, predict the remaining parameters needed to define a parallel-jaw grasp




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Sundermeyer, Martin, et al. "Contact-graspnet: Efficient 6-dof grasp generation in cluttered scenes." *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2021.

# 6-DoF Grasping: AnyGrasp

**Practical modern system:** pretrained 6-DoF grasp detection and tracking in cluttered scenes.






# Modern 6-DoF Grasp Generation: Diffusion Models

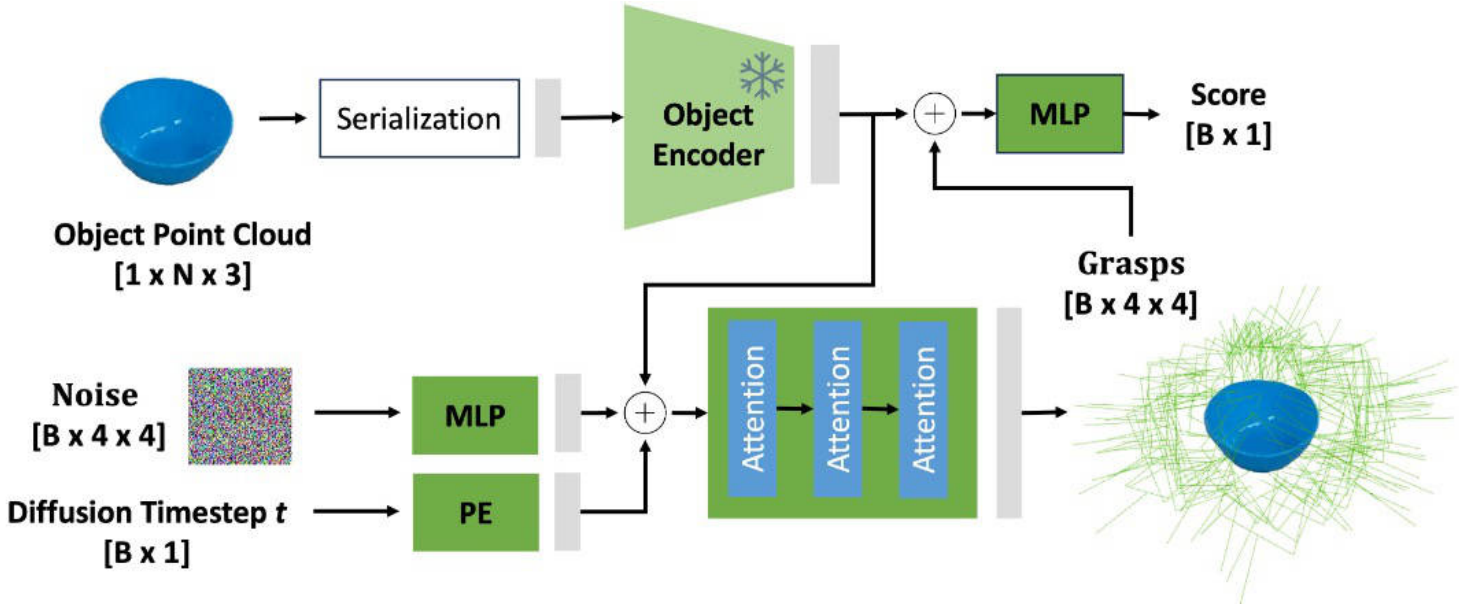
**Key idea: an object often has many valid grasps, not one correct answer.**

- Learn a distribution over feasible 6-DoF grasp poses
- Start from random/noisy grasp poses and iteratively refine them toward plausible grasps
- Can generate multiple diverse grasp candidates
- Particularly well suited to the multi-modal nature of grasping
- Candidates can then be ranked or filtered using grasp quality / collision checks

- **Typical Pipeline:** Point cloud / RGB-D → Diffusion-based grasp generator → → many candidate 6-DoF grasps → score / filter → execute selected grasp

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# GraspGen




arXiv:2507.13097v1 [cs.RO] 17 Jul 2025

NVIDIA

2025-7-18

## GraspGen: A Diffusion-based Framework for 6-DOF Grasping with On-Generator Training

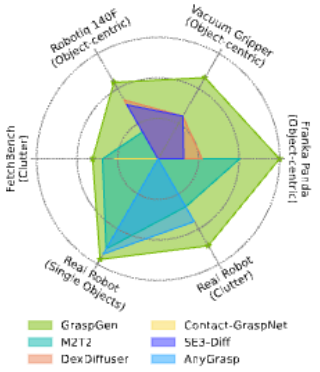
Adithyavairavan Murali Balakumar Sundaralingam Yu-Wei Chao Wentao Yuan* Jun Yamada*
Mark Carlson Fabio Ramos Stan Birchfield Dieter Fox* Clemens Eppner*
* Work done at NVIDIA

https://graspgen.github.io

### Abstract

**Grasping is a fundamental robot skill, yet despite significant research advancements, learning-based 6-DOF grasping approaches are still not turnkey and struggle to generalize across different embodiments and in-the-wild settings. We build upon the recent success on modeling the object-centric grasp generation process as an iterative diffusion process. Our proposed framework, *GraspGen*, consists of a Diffusion-Transformer architecture that enhances grasp generation, paired with an efficient discriminator to score and filter sampled grasps. We introduce a novel and performant on-generator training recipe for the discriminator. To scale GraspGen to both objects and grippers, we release a new simulated dataset consisting of over 53 million grasps. We demonstrate that GraspGen outperforms prior methods in simulations with singulated objects across different grippers, achieves state-of-the-art performance on the FetchBench grasping benchmark, and performs well on a real robot with noisy visual observations.**

### 1. Introduction




Robot grasping has seen significant advances in recent years: including data generation (Eppner et al., 2021), generalization across embodiments (Xu et al., 2021), integration with touch sensing (Calandra et al., 2018; Murali et al., 2018), operating in complex cluttered environments (Murali et al., 2020b), language prompting (Murali et al., 2020a; Tang et al., 2023), and real-world RL algorithms (Kalashnikov et al., 2018).

However, recent results show that critical gaps still exist in the development of a general-purpose grasping system. In the FetchBench (Han et al., 2024) benchmark, state-of-the-art (SOTA) grasping systems achieve sub-20% accuracies. Similarly, the OK-Robot effort (Liu et al., 2024), which introduced a knowledge-based system for mobile manipulation in-the-wild, reported a notable error rate of 8% (30 errors out of 375 trials) due to grasp model failures alone. These grasping models perform at approximately 60% accuracy in their evaluations. The evaluation of Robo-ABC (Ju et al., 2025) (which uses a SOTA grasp method) as a baseline in RAM (Kuang et al., 2024) shows sub-50% success rates. This highlights the need for further advancements in grasping frameworks to ensure their reliability as subroutines in higher-level reasoning systems (Dalal et al., 2024; Deshpande et al., 2025; Huang et al., 2024b; Liu et al., 2024).

Furthermore, grasping systems are not yet turnkey, and making them more flexible remains an open systems research challenge. For instance, classical model-based approaches to grasp generation required precise object pose information (Deng et al., 2020) which does not generalize for the unknown object setting. Other methods necessitate multi-view scans for a single object (Lum et al., 2024), making them impractical for cluttered

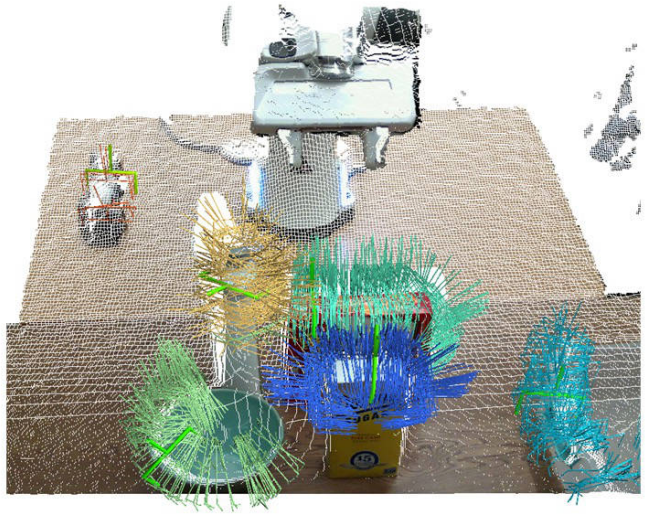
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# Beyond Geometry: Semantic and Task-Aware Grasping

**Geometry-only grasping – Where can I grasp?**

Input: RGB-D

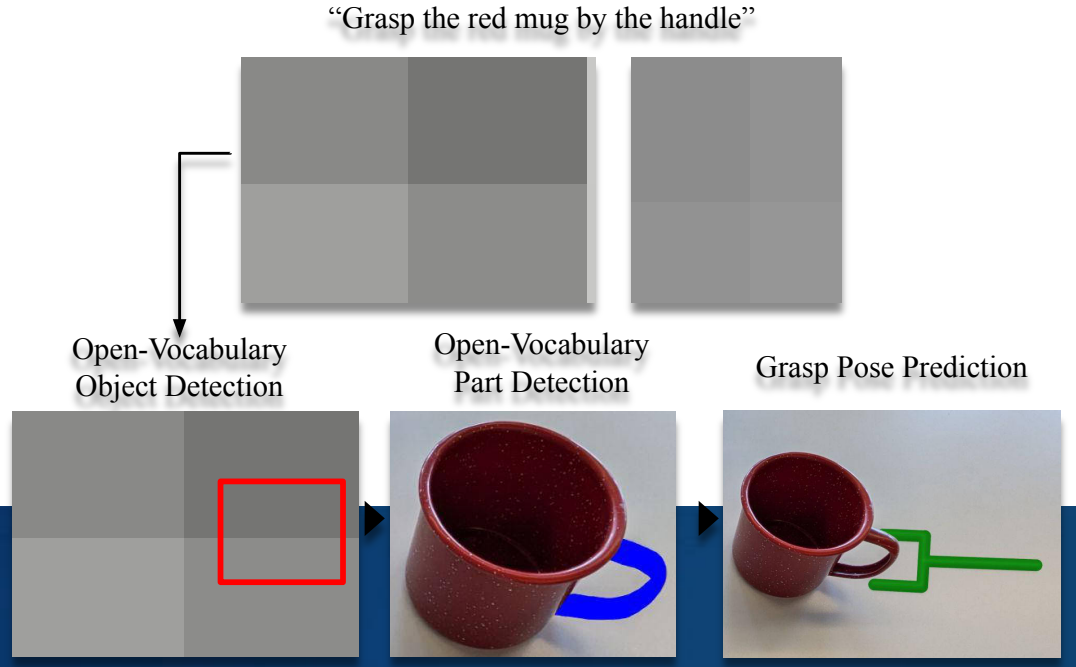
Output: feasible 6-DoF grasp poses



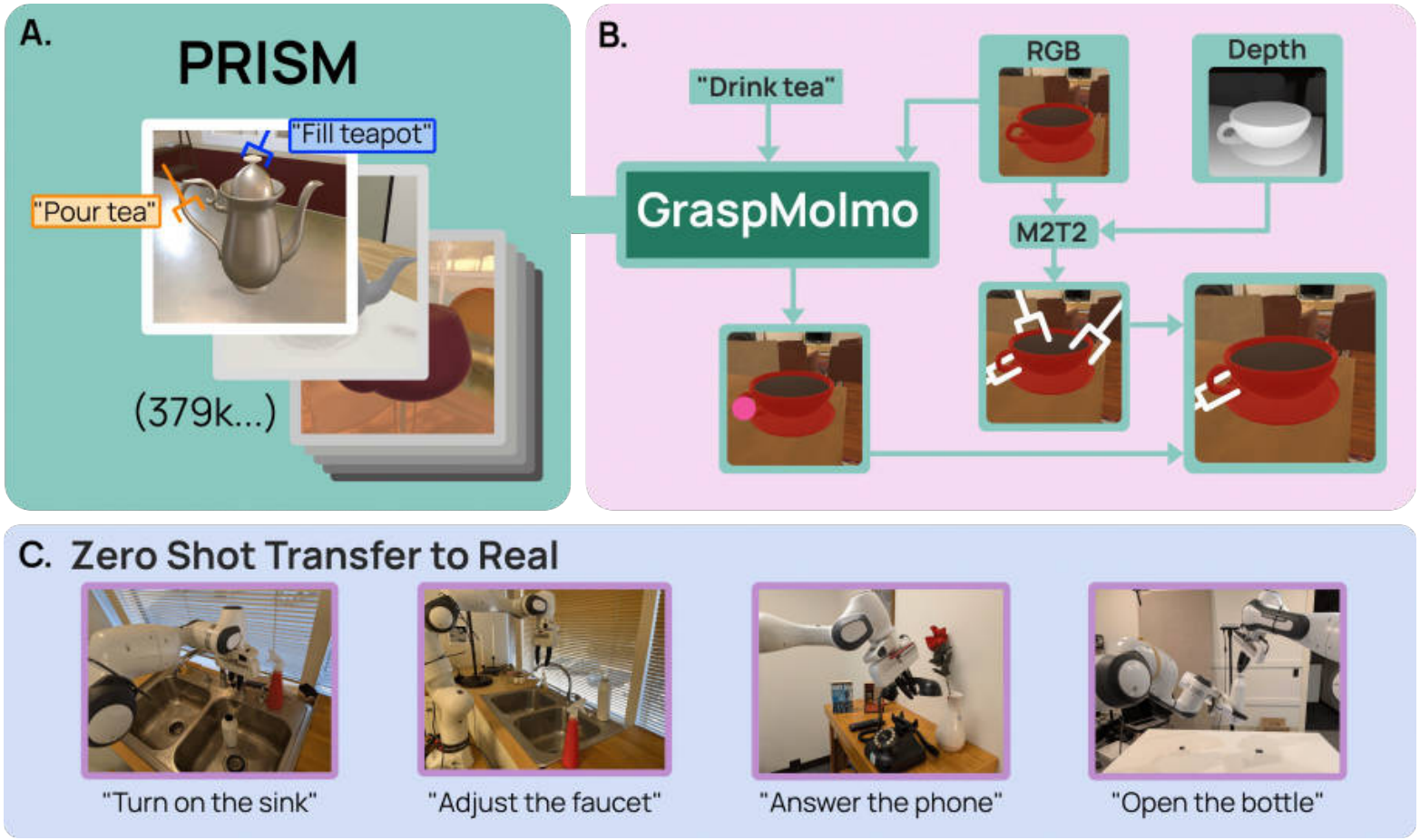
**Semantic / language-conditioned grasping – *What* should I grasp, and *how* should I grasp it for the task?**

Input: RGB-D plus language instruction

Output: task-conditioned 6-DoF grasp pose(s)



# Example: GraspMolmo




arXiv:2505.13441v3 [cs.RO] 12 Sep 2025

## GraspMolmo: Generalizable Task-Oriented Grasping via Large-Scale Synthetic Data Generation

**Abhay Deshpande**[1], **Yuquan Deng**[1], **Arijit Ray**[2], **Jordi Salvador**[1], **Winson Han**[1], **Jiafei Duan**[1,3], **Kuo-Hao Zeng**[1], **Yuke Zhu**[4], **Ranjay Krishna**[1,3], **Rose Hendrix**[1]
[1]PRIOR @ Allen Institute for AI, [2]Boston University,
[3]University of Washington, [4]University of Texas at Austin
Corresponding Author: abhayd@allenai.org

**Abstract:** We present GraspMolmo, a generalizable open-vocabulary task-oriented grasping (TOG) model. GraspMolmo predicts semantically appropriate, stable grasps conditioned on a natural language instruction and a single RGB-D frame. For instance, given "pour me some tea," GraspMolmo selects a grasp on a teapot handle rather than its body. Unlike prior TOG methods, which are limited by small datasets, simplistic language, and unrealistically simple scenes, GraspMolmo learns from PRISM, a novel large-scale synthetic dataset of 379k samples featuring complex environments and diverse, realistic task descriptions. We fine-tune the Molmo vision-language model on this data, enabling GraspMolmo to generalize to novel open-vocabulary instructions and objects. In challenging real-world evaluations, GraspMolmo achieves state-of-the-art results, with a 70% prediction success on complex tasks, compared to the 35% achieved by the next best alternative. GraspMolmo also successfully demonstrates the ability to predict semantically correct bimanual grasps zero-shot. We release our synthetic dataset, code, model, and benchmarks to accelerate research in task-semantic robotic manipulation, which, along with videos, are available at this URL.

**Keywords:** Robots, Learning, Task-Oriented Grasping, Manipulation

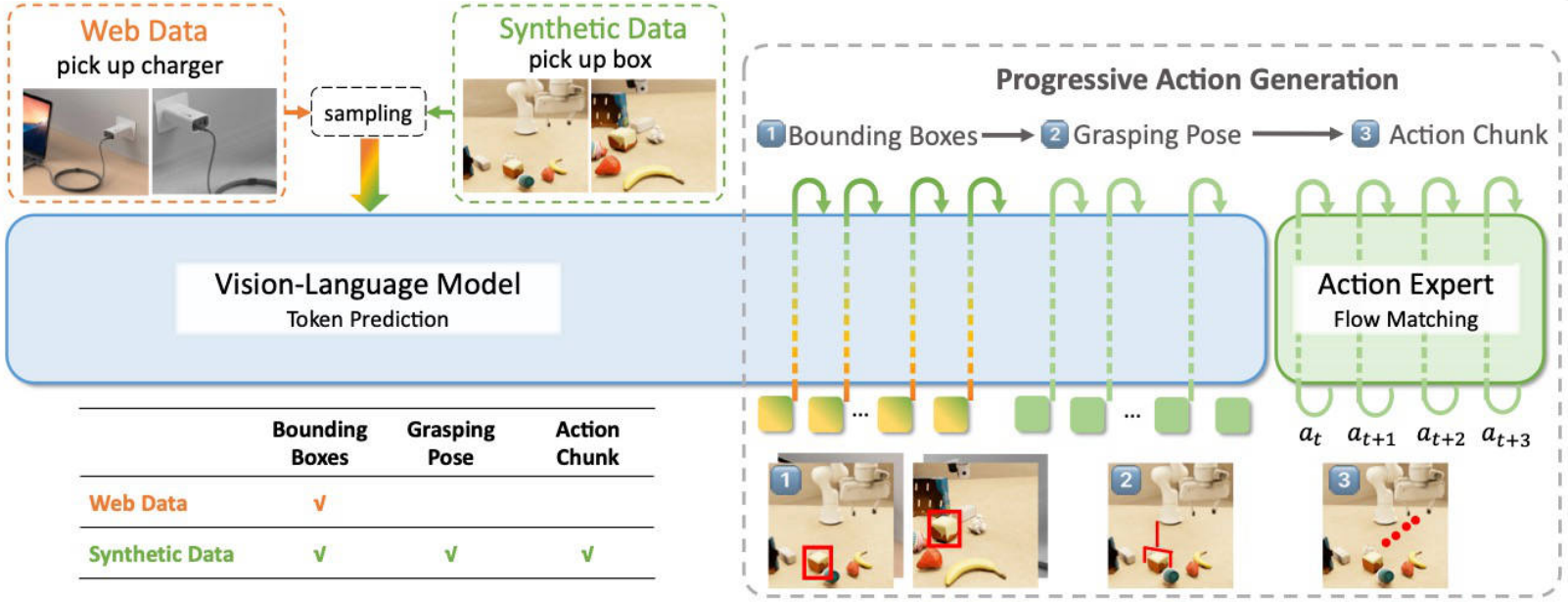
### 1 Introduction

Robotic grasping has progressed significantly, with models now capable of robustly predicting stable grasps for a wide range of objects. Yet most existing methods operate in an object-centric fashion: they predict grasps that are stable for a given object but oblivious to the task at hand. In the real world, however, how an object should be grasped depends on what the robot is trying to do with it.

Consider a kitchen knife. A task-agnostic grasping system might simply choose a stable grasp anywhere on the knife, including the blade. While the blade might be an appropriate grasp site when handing the knife safely to another person, it would be ineffective for the purposes of chopping vegetables. These task-specific differences highlight the need for task-oriented grasping (TOG): The same physical object requires dramatically different grasps depending on the intended task [1, 2].

Recent work has introduced relevant datasets and benchmarks [2, 3, 4], but they are limited in realism and diversity. Some contain relatively simple scenes without any distractors and lack diverse object categories [2]. Others often templatize (e.g., "grasp the mug to pour" or "grasp the pan to clean"). These simplifications fail to capture the complexity of real-world environments and the wide variety of language humans use to specify tasks, such as "mince some garlic" or "do the dishes". Furthermore, many models require pre-segmented point clouds [4] or multi-view observations [3], restricting real-time deployment.

# Example: GraspVLA




| | Bounding Boxes | Grasping Pose | Action Chunk |
|---|---|---|---|
| Web Data | √ | | |
| Synthetic Data | √ | √ | √ |

arXiv:2505.03233v3 [cs.RO] 27 Aug 2025

**GraspVLA: a Grasping Foundation Model Pre-trained on Billion-scale Synthetic Action Data**

Shengliang Deng*,1,3 Mi Yan*,1,2 Songlin Wei1,2 Haixin Ma1 Yuxin Yang1 Jiayi Chen1,2 Zhiqi Zhang1,2
Taoyu Yang2 Xuheng Zhang2 Wenhao Zhang2 Heming Cui3 Zhizheng Zhang†,1,4 He Wang†,1,2,4

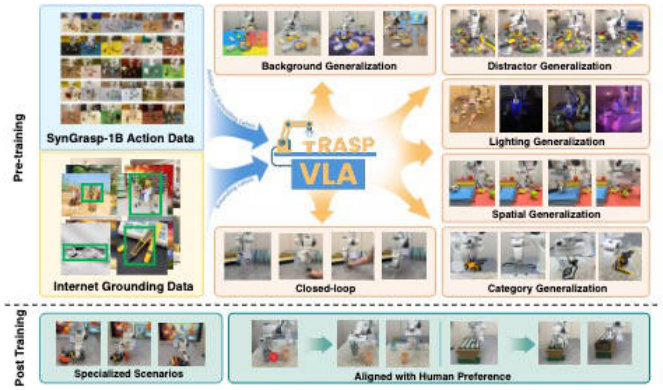



Figure 1: GraspVLA is a grasping foundation model pre-trained exclusively on billion-scale synthetic action data and co-trained with Internet semantics data. It exhibits direct sim-to-real transfer and strong zero-shot generalization across diverse aspects, as well as few-shot adaptability to specialized scenarios and human preferences.

**Abstract:** Embodied foundation models are gaining increasing attention for their zero-shot generalization, scalability, and adaptability to new tasks through few-shot post-training. However, existing models rely heavily on real-world data, which is costly and labor-intensive to collect. Synthetic data offers a cost-effective alternative, yet its potential remains largely underexplored. To bridge this gap, we explore the feasibility of training Vision-Language-Action (VLA) models entirely with large-scale synthetic action data. We curate SynGrasp-1B, a billion-frame robotic grasping dataset generated in simulation with photorealistic rendering and extensive domain randomization. Building on this, we present GraspVLA, a VLA model pretrained on large-scale synthetic action data as a foundational model for grasping tasks. GraspVLA integrates autoregressive perception tasks and flow-matching-based action generation into a unified Chain-of-Thought process, enabling joint training on synthetic action data and Internet semantics data. This design helps mitigate sim-to-real gaps and facilitates the transfer of learned actions to a broader range of Internet-covered objects, achieving open-vocabulary generalization in grasping. Extensive evaluations across real-world and simulation benchmarks demonstrate GraspVLA's advanced zero-shot generalizability and few-shot adaptability to specific human preferences. We will release SynGrasp-

* Equal contribution with the order determined by rolling dice. † denotes corresponding authors.
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[1]Galbot, [2]Peking University, [3]The University of Hong Kong, [4]Beijing Academy of Artificial Intelligence

# Which Model Should I Start With?

- **AnyGrasp** – practical, pretrained, robust grasp detection/tracking
- **GraspMolmo / GraspVLA:** language-conditioned, semantic/task-aware grasping: choose what to grasp and how to grasp it for the task

- **Contact-GraspNet** – excellent model for understanding how modern 6-DoF grasp generation works
- **GraspNet-1Billion** – key dataset and benchmark ecosystem

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# Summary

# Segmentation

- **Semantic segmentation:** assign a semantic class to every pixel
- **Instance segmentation:** predict separate masks for individual object instances
- For *C* **mutually exclusive** classes: predict *C* logits per pixel and apply softmax
- Encoder–decoder networks recover spatial resolution; **skip connections** preserve fine spatial detail and object boundaries
- Modern models include transformer-based and promptable/open-vocabulary segmentation

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# Grasp Prediction

- **Planar grasps:** image-plane position, orientation, width and quality
- **6-DoF grasps:** 3D position + 3D orientation, usually plus gripper width and quality
- Modern models directly generate candidate grasps from RGB-D / point clouds
- Semantic/task-aware models additionally use language to decide **what** to grasp and **how**

CRICOS No.00213J